Does Money Move Votes in Washington?

An identification failure, measured: an exceptionally strong money–performance association that no available observational decomposition can separate from selection

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AI-assisted drafting and analysis review. All reported statistical summaries can be recomputed from public code and the pinned derived panels cited below. The model-derived residuals themselves cannot presently be independently regenerated, because the forecast implementation is not public; the pinned panels carry the residuals so that every slope, interval and correlation — the contestable part — is recomputable without it. The paper source, code, and data-acquisition recipe are public at https://github.com/skirby359/who-decides. Contact: kirby@tikorconsulting.com.

DRAFT — pending human/editorial sign-off. `scripts/verify_money_votes.py` *scrapes this paper and asserts its figures against the data, with the exceptions the script names in its own output; that gate is automated and is not the sign-off. The sign-off is a person reading the paper end to end, recorded in* `money-votes-submission-notes.md` *§Sign-off.*

Paper #4 of the electoral-health series (companion to `who-decides-washington.md`, `safe-seat-washington.md`, `donor-class-and-the-electorate.md`, *and* `cross-state-fec-money.md`*).*

Abstract

The three preceding papers in this series establish who votes, whether their vote decides anything, and who funds the candidates. This paper asks the question those invite — does the money change the result? — and reports that the available public record cannot answer it, while characterizing precisely *why* it cannot: treatment is endogenous, sparse, strategically assigned, and observed through several non-equivalent channels. Across 163 Washington legislative and congressional races from 2018 to 2024, the ratio of Democratic to Republican fundraising correlates with candidate overperformance at **+0.60**, the strongest of any measured factor, ahead of incumbency (+0.43) and candidate quality (+0.34) — an ordering that holds when every factor is recomputed on the common finance-complete sample. The paper then examines **three different observable manifestations of political money** — candidate receipts, the Democratic campaign's spending *allocation*, and outside groups' independent expenditure — and none supplies identification of the causal effect the title asks about; they are different treatments, not three replications of one null. The allocation cut is exploratory: it measures the mix of one side's spending, not its level, and its squared holdout correlation of essentially **zero** cannot rule out an effect of having more money. The directional test — the observational OLS of fundamentals-net residual on net independent-expenditure advantage — is estimable across five cycles and thirty-four scorable Washington congressional races, and returns **+0.5 points of residual per $1M of net pro-Democratic IE with a case-resampling bootstrap interval spanning −0.6 to +2.8**. That interval is an interval for the observational slope, not for a causal effect, which confounding can place outside it; and the slope itself is a single-district leverage result — deleting one race (WA-08 2018, Cook's distance 0.69) moves it to **−0.04**, and a district-clustered bootstrap widens the interval to **−1.6 to +1.3**. Extending the same estimator to Washington's state legislative races — **$51.7M of PDC independent expenditure across 129 scorable district-cycles, direction-coded on 100% of rows** — multiplies the cells but not the usable treatment variation: every interval still spans zero, and the sign of the point estimate is **specification-dependent**, running from −3.8 to +4.9 depending on whether electioneering communication is counted as advocacy and whether money is matched to the contest it named. The most heavily funded race in the federal panel, Washington's 3rd in 2024, finished 0.06 points from its fundamentals. The honest verdict: money in Washington is clearly a **marker** of candidate strength; whether it is *also* a mover of votes is not established either way, because every available observational decomposition fails to separate selection into spending from an effect of spending. The failure is one of identification, not merely of data quantity.

Keywords: campaign spending; independent expenditures; electoral effects; endogeneity; null results; data availability; state legislative elections; Washington.

The question, and why it resists an answer

The preceding papers in this series each end at the same edge. The turnout paper shows an older, smaller electorate decides off-cycle races. The safe-seat paper shows most general elections are settled before November. The donor paper shows the population funding those races is not representative of the electorate voting in them — donors skew sharply older, more geographically concentrated and, where party is observable, more Democratic, while being *more* likely to vote, not less. Each invites the obvious next question: **so does the money actually change who wins?**

It is the hardest question in the series, and the reason is structural rather than technical. Campaign money is *endogenous to candidate strength*. Strong candidates attract money; money then appears alongside strong results. Donors and independent-expenditure committees also behave strategically, moving resources toward races they judge close and toward candidates they judge threatened — which can invert the naive correlation entirely, making spending look harmful. Untangling this is a fifty-year-old problem in political science (Jacobson 1978), and the credible answers come from designs this data does not support: instrumental variables (Gerber 1998), repeat-challenger panels (Levitt 1994), or randomized field experiments (Kalla & Broockman 2018).

This paper does not solve it. What it does is report what Washington's public record shows, run the tests that record permits, and state precisely what those tests do and do not estimate. **The contribution is the identification failure itself, characterized:** the observational association is exceptionally strong, its estimate is imprecise and leverage-dependent, and even a perfectly estimated association would not identify the causal effect, because nothing in this record separates selection into spending from an effect of spending. A reader looking for a verdict on whether money buys elections will not find one here, and should be suspicious of anyone offering one from this evidence base — including from an earlier draft of this paper, whose directional estimate carried the opposite sign on a seventh of the data.

Finding 1 — Money is the single strongest correlate of overperformance

The starting observation is not subtle. Across the 163 baseline-scorable Washington legislative and congressional races from 2018 to 2024 (`scripts/diag_overperformance_patterns.py`), *overperformance* — the actual Democratic two-party share minus the district's fundamentals-based baseline — correlates with the fundraising ratio more strongly than with anything else measured. Because finance is present for only **128** of the 163 cells, the competing correlations are reported twice: on each factor's full available sample, and on the common finance-complete sample, so the ranking is not an artifact of the factors seeing different races:

factor	full sample (n=163)	common sample (n=128)
fundraising, log2(D receipts / R receipts)	**+0.60** *(n=128)*	**+0.60**
incumbency	+0.43	+0.41
candidate quality index	+0.34	+0.36
local trend	+0.31	+0.30
midterm year	≈0	+0.13

Fundraising leads on either basis. The relationship is monotonic, not driven by a tail. Sorting races by who out-raised whom: where the Democrat out-raised the Republican, the Democrat beat the district baseline by an average of **+4.22** points; where funding was even, **+2.11**; where the Republican out-raised, **−1.77**.

> **The cell frame is pinned, and it has moved twice — each time deliberately, and each time recorded here.** The fundraising feature is matched against `candidate_finance`, a live table: the frame behind the first draft held **109** both-side cells and gave +0.55, +4.20, +2.37 and −1.93; loading more filings took it to 129 and +0.58. An external referee then caught the linkage itself resolving candidates by `(cycle, last name, first initial)` alone, pooled across all four office codes — and the collision audit (`scripts/diag_finance_linkage_audit.py`) found 2 of 326 lookups attaching the wrong records, one of them crediting a **different candidate of the same name** in another chamber with $190K his namesake never raised. The linkage is now scoped to the cell's own office and district, which removes that cell's finance pair entirely (**128** both-side cells) and moves the correlation to **+0.60**. The universe is unchanged at 163 baseline-scorable cells, and so is every conclusion — fundraising remains the largest measured factor on either sample. The frame is frozen at `docs/reference/overperformance_cells_2026-08-15.csv` (its predecessor is retained for audit), which is what `scripts/verify_money_votes.py` asserts against, so a reader re-running the derivation gets the figures this paper prints rather than whatever the table holds that week.

Taken alone this looks like a straightforward answer, and it is the number a campaign consultant would quote. The rest of this paper is about why it is not one.

Finding 2 — Three manifestations of money, and what each can and cannot test

Candidate receipts, a campaign's own spending allocation, and outside groups' independent expenditure are **three different treatments**, not three measurements of one. A causal effect of candidate spending can coexist with a null for independent expenditure, and vice versa. This section examines each manifestation on its own terms and states what its test does and does not estimate; none of the three supplies identification of the causal effect the title asks about, and they are not presented as three replications of one null.

2a. Spending composition carries no incremental predictive information — an exploratory cut, not a causal test

If spending buys votes, *how* it is spent might matter — a campaign that puts its money into field organizing might perform differently from one that puts it into television. Washington's Public Disclosure Commission publishes itemized expenditures with a purpose code per transaction (Socrata `tijg-9zyp`; 272,320 candidate rows 2018–2024, $286M, ~62% of dollars carrying a code). From these, `scripts/diag_expenditures_vs_residual.py` derives each candidate's field, media, and professional *shares* of coded operational spend, and tests them against the residual.

Two design limits are stated before any number, because they are what demote this cut to exploratory. **It measures only the Democratic campaign's allocation** — if Republican field spending moves Democratic vote share, this test cannot see it; a race-level causal design would need both campaigns, or their difference. And **allocation mix and spending level are different treatments**: a world where money has a substantial causal effect but campaigns optimally converge on similar mixes produces exactly the zero correlations below. What this cut can test is narrow — whether spending *composition* carries predictive information the core features lack — and that is all it is offered as.

The shares correlate with the residual at essentially zero — field **+0.02**, media **+0.04**, professional **−0.05**. Total spend correlates at +0.23, but that is the fundraising scale signal from Finding 1 arriving again, not allocation. The sharper cut is cross-cycle holdout: fit on 2022, predict 2024. The first metric below is the **squared holdout correlation** — the square of the Pearson correlation between predictions and holdout outcomes. An earlier draft called this "holdout R^2", which it is not: squaring discards sign, and three of the four holdout correlations are wrong-signed (r = +0.02, −0.12, −0.18, −0.17), so the squares flatter the models. The standard **predictive R^2** (1 − SSE/SST) is therefore reported beside it, and it is negative for every specification — the models predict *worse than the holdout mean*.

model	squared holdout correlation	predictive R^2
core candidate-quality features	0.000	−1.09
core + field share	0.015	−2.75
core + all allocation shares	0.031	−3.00
allocation shares alone	0.028	−0.18

In-sample squared correlation rises from 0.037 to 0.101 as shares are added while the holdout numbers worsen — the signature of overfitting, not signal. One leakage caveat is owed on the word "holdout": the target residual comes from the forecast model, whose coefficients were tuned on the full residual matrix, so 2024 outcomes participated in constructing the variable 2024 is here held out on. Allocation itself never enters that first stage, but a fully clean version would fit the residualization on 2022 alone — one more reason this cut is exploratory rather than a causal null.

> **This block is pinned too, and eight of its figures moved when it was (2026-08-01).** The allocation frame is derived against the forecast model's *residual*, so it shifts whenever the model does — and the model has moved since this section was written, while the fundamentals baseline behind Finding 1 has not. That is why the two figures computed against `overperf` rather than `residual` are unchanged (field-vs-overperformance −0.24, and both persistence coefficients) and every figure computed against the residual moved: media +0.05 → +0.04, professional −0.03 → −0.05, total spend +0.26 → +0.23, and the four holdout cells with the in-sample pair. The 2026-08-15 linkage correction re-pinned the residuals a second time, moving the holdout cells again by similar amounts (the allocation-alone cell has run 0.041 → 0.022 → 0.028 across the three bases) while every correlation held at its printed precision. **Nothing in the finding changes on any basis** — the holdout numbers stay at the floor, most holdout correlations are wrong-signed, and the shares carry no out-of-sample information. The frame is frozen at `docs/reference/expenditures_vs_residual_2026-08-15.csv`, which is what `scripts/verify_money_votes.py` asserts the correlations against. Notably, allocation *is* a real and stable candidate trait: field-share persistence across cycles runs r = 0.83 and media-share r = 0.998. It is a stable feature that carries no information about the outcome. And the one directional hint runs against the folk theory: field share correlates **−0.24** with raw overperformance, so more ground spending is associated with slightly worse results, not better.

2b. The forecast model discards the money term — a forecasting result, not a causal falsification

A second observation comes from an unusual direction: the forecasting model this series relies on was tuned without reference to this question, and it independently concluded that fundraising adds no *predictive* information once the baseline is known. What that does and does not show is stated at the end of this subsection, because an earlier framing over-claimed it.

In Washington's post-2022 districts, where the redistricting boundary filter collapses history to a single post-redistricting cycle, the model **zeroes its fundraising-advantage term entirely**. The reason is that the single-cycle baseline already absorbs it: the incumbent ran in that cycle, their fundraising advantage manifested in that result, and adding the term on top double-counts. Before this correction the model predicted a D+28 result in a district whose incumbent's actual history ran D+5 to D+8. The term was not dropped for ideological reasons but because leaving it in produced worse forecasts.

What this establishes is a **forecasting result**: fundraising adds no incremental predictive information conditional on the district's own recent result. What it cannot establish is anything causal, because conditioning on a prior outcome absorbs both signal and treatment. Two worlds are observationally identical here: one where fundraising is merely a marker whose information was already present in past results, and one where fundraising **caused** part of the previous result, so its causal effect is now embedded in the lagged outcome the model conditions on. An earlier draft counted this as one of three causal nulls; it is not one, and it is not offered as evidence against a money effect.

2c. The directional test runs, and cannot sign the effect

The third test is the closest thing to a directional design the data permits. `scripts/diag_ie_vs_margin.py` regresses each race's *fundamentals-net residual* — actual minus model-predicted Democratic share, not the raw margin — on the net pro-Democratic independent-expenditure advantage. Using the residual rather than the margin is what makes this a test of money's marginal effect rather than a restatement of district partisanship.

Across the 34 scorable district-cycles the slope is **+0.515 points of residual per $1M net pro-Democratic IE, with a 95% bootstrap interval of −0.600 to +2.821 and Pearson r = +0.186**. Read that interval for what it is: **an interval for the observational OLS slope, not for the causal effect of $1M of IE.** Under uncontrolled confounding the causal effect can lie outside it, and even an infinitely narrow interval around zero would not identify a zero causal effect — a narrower interval around a better-estimated association is still an association. Within its own terms the interval crosses zero, so no association can be asserted; it is also wide enough to be compatible with associations that would decide a close race. The point estimate is not a finding, and neither, on its own, is the width: the design is what prevents a stronger statement.

The slope is a single-district leverage result, and that is reported rather than smoothed over. The leave-one-out sweep this paper already applies to its placebo and legislative specifications, applied to this headline regression, runs from **−0.035** (dropping WA-08 2018) to **+0.832** (dropping WA-03 2024): deleting one race erases the positive slope entirely. WA-08 2018 pairs the panel's largest net pro-Democratic IE (+$8.26M) with a +8.78-point residual and carries a Cook's distance of **0.69** in a 34-observation regression; dropping every WA-08 observation gives **−0.065**. The case-resampling interval also treats race-cycles as iid while the panel measures the same ten districts repeatedly — in tension with this paper's own placebo discussion of persistent district traits — so a **district-clustered bootstrap** (10 clusters) is reported beside it: **−1.595 to +1.268**; clustering on era × district across the 2022 redistricting boundary (20 clusters) gives −1.016 to +1.390. Every version spans zero, and the clustered intervals are materially wider. On a standardized treatment scale the slope is **+0.93 points per one standard deviation of net IE** (SD = $1.80M in this panel) — the scale that makes the legislative comparison below meaningful.

The sign of the point estimate should be treated with particular caution, because it is not stable. An earlier version of this analysis, run on 2024 alone, reported a *negative* slope of −0.39 (Pearson r = −0.39, n = 7) and read it as the textbook endogeneity signature: outside money flowing toward the side that is struggling. Extending the panel to five cycles reversed the sign. Both readings are consistent with the same underlying truth, which is that at these sample sizes the sign is a coin flip and the interval is the only honest summary. Endogeneity remains the correct prior — spending is targeted at expected closeness, not assigned at random — and it is precisely why a signed estimate from this design would be uninterpretable even if the interval excluded zero.

The single most instructive case is the most heavily funded. **Washington's 3rd Congressional District in 2024 attracted $18.61M in total independent expenditure — the largest in the panel, and 22nd nationally among the 387 U.S. House races drawing any — with a net $6.09M advantage on the Democratic side. It finished 0.06 points from its fundamentals-based prediction.** Eighteen million dollars of outside money, and the race landed almost exactly where the district's underlying partisanship said it would. That is a single observation and proves nothing on its own, but it is a striking one.

(A previous draft of this section called WA-03 "the most IE-saturated House race in the country" at $40.1M. Both halves were

artifacts of a data defect: FEC reports most independent expenditures twice, once as a 24/48-hour notice and again on the periodic Schedule E, and the loader summed both. At the true $18.61M the race ranks 22nd; at the doubled $40.1M no race in the country exceeded it, which is what manufactured the superlative. The defect and its correction are documented in the data section below.)

Finding 3 — The test runs now, and is still underpowered

The design in 2c is the right one, and until recently it could not be run at all. What has changed is worth stating precisely, because the limitation that remains is different in kind from the one it replaced.

Directional independent-expenditure data now spans five cycles. FEC Schedule E carries a support-or-oppose flag and a district, which is what makes a directional test possible at all. The Washington U.S. House panel holds **2,215 flagged rows across all ten districts and $75.7M**, distributed as follows:

cycle	flagged rows	total IE	scorable races
2018	887	$19.73M	8
2020	374	$5.46M	9
2022	460	$25.06M	10
2024	**444**	**$25.4M**	7
2026	**50**	**$98K**	0

2026 is a partial current-cycle trickle with no national-environment data to net a residual against and no result to net it from, so it contributes nothing; the four completed cycles yield **34 scorable Washington House races**, uncontested districts dropping out.

Three of those four cycles were absent from an earlier version of this paper, which reported a single scorable cycle and seven races and called that limitation its most citable result. The backfill that removed the limitation is the one the earlier draft itself prescribed. It took about five minutes of API time — the cost was never the obstacle, and saying otherwise was the draft's own error. What made the earlier figures wrong was not the missing cycles but a de-duplication defect described in the data section below, which inflated every independent-expenditure total by roughly a factor of two.

The state-legislative money enters the test too, and it does not rescue it. Washington's PDC records **$51,723,243.45** of independent expenditure identifying a legislative candidate across 2018–2024, on **4,653 filed rows carrying a support/oppose direction, a named candidate, a filer id and a jurisdiction on 100% of them**. Matched to the same fundamentals-net residual, that yields **129 scorable district-cycles** — close to four times the 34-race federal panel — and the estimate remains uninformative:

specification	n	cells with ≥$25K	slope	95% bootstrap	district-cluster 95%	Pearson r
express advocacy, race-matched	129	17	**+4.871**	−29.126 to +24.941	−38.251 to +14.109	+0.085
express advocacy, district-aggregate	129	53	−2.043	−11.170 to +3.203	−11.278 to +2.816	−0.053
all directional, race-matched	129	28	−3.836	−15.619 to +3.664	−18.890 to +3.225	−0.083
all directional, district-aggregate	129	60	+0.107	−5.090 to +4.790	−6.523 to +5.770	+0.004

Every specification now runs on the full 129-cell scorable universe: a scorable race that attracted no race-matched IE enters with net IE = 0 rather than dropping out. An earlier version dropped those cells from the race-matched rows (n = 127), treating zero treatment as missing data; the correction, prompted by an external referee, moved the primary slope from +4.890 to +4.871 and no conclusion. The district-cluster column resamples the 37 legislative districts rather than individual race-cycles, for the same reason as the federal panel's clustered interval.

Every interval spans zero and the sign is specification-dependent. Two of the four reverse sign on the deletion of a single race. Multiplying the cells by nearly four did not narrow the answer, and the reason is worth stating exactly, because "the sample was too small" is not it: **raw cell count is not the binding quantity; usable variation in treatment is.** Only 13–47% of cells attract any material independent expenditure, so the slope is identified by a few dozen contests whichever specification is chosen — and those are exactly the contests money selected for being close. The panels are also not comparable on raw interval width: a $1M x-unit is an ordinary congressional IE campaign and an enormous intervention in a legislative contest. On the standardized scale, one standard deviation of net IE moves the point estimate **+0.93** points in the federal panel (SD $1.80M) against **+0.33** points in the primary legislative specification (SD $0.067M) — with every interval, standardized or not, spanning zero.

Two properties of the PDC record decide the specification and are not incidental. **Three quarters of the directional dollars are electioneering communication, not express advocacy** — $38.70M against $13.03M — and the two run in opposite directions, express advocacy 78% *for* candidates and electioneering 61% *against* them. Pooling them inverts the panel's directional balance, which is why the two are reported apart rather than summed. And the filing identifies a candidate's *chamber* but not their House position, while the outcome is a single contest; the race-matched rows count only money naming a candidate on the ballot in the scored race, and the district-aggregate rows show what attributing all three seats' money to one seat's residual does instead.

(An earlier version of this paper reported that this money "carries no support/oppose flag" on all but 5 of 4,456 rows and "cannot enter a directional test at all", and treated that as a limitation of Washington's disclosure regime. It was a limitation of our own extraction. The PDC publishes direction in the C6.3 "Identified Entities" section of form C-6, in the same dataset the loader was already reading and had been reading all along; the loader consumed only the C6.2 itemized-expenditure section. The correction is documented in `pdc-c6-direction-audit.md`*, and the ingest is* `load_pdc_ie_targets`*. The conclusion this paper draws is unchanged — the point estimate is still unsignable — but the reason is different in kind, and the earlier reason was wrong.)*

The data defect, and why it is reported rather than quietly fixed. FEC files most independent expenditures **twice** — once as a 24- or 48-hour notice on Form 24, then again on the committee's periodic Form 3X Schedule E — under different transaction ids. Nothing about the pair looks duplicated. An earlier loader summed both, which roughly doubled every total in this paper: WA-03's 2024 IE was reported as $40.1M against a true $18.61M, and the same factor applied to every district and cycle. It was caught by reconciling against FEC's own per-candidate aggregate, which returned almost exactly half the loaded figure for every Washington House candidate checked.

Two consequences deserve stating plainly. The inflated figure is what produced the claim that WA-03 was the most IE-saturated House race in the country: at $40.1M nothing exceeded it, and at the correct $18.61M twenty-one races do. And a defect of this shape is invisible to internal consistency checks, because every figure derived from the contaminated table agreed with every other. Only an external source could catch it. `scripts/diag_fec_ie_bulk_crosscheck.py` is that check, and it needs no API key: it reconciles the loaded totals against FEC's public bulk files, so anyone reproducing this work can verify the independent-expenditure figures without credentials. All five cycles reconcile to those files exactly.

The two secondary tests now run, and neither discriminates

An earlier draft named an early-versus-late spending split and a next-cycle placebo as the tests a multi-cycle panel would unlock. Both were run (`scripts/diag_ie_early_late.py` , `scripts/diag_ie_next_cycle_placebo.py`). Both return nothing that survives inspection, and the *reason* they fail has changed: it is no longer the number of cycles but the number of races within them that attract money at all.

Early versus late. Splitting each race's independent expenditure at 30 days before election day and regressing the residual on both windows jointly gives **−1.128 points per $1M of early money and +2.129 of late money**, both intervals spanning zero, R^2 0.083. Late exceeds early at every cutoff tried (14, 30 and 60 days), which is the one consistent pattern — but the early coefficient changes sign across those cutoffs, and at 60 days the two nearly converge, which is what a split carrying no information looks like as the window widens toward the pooled total. The deeper problem is identification: of 34 scorable races, **15 attracted any material independent expenditure and only 11 attracted it in both windows**, with early and late spending correlated at +0.753 among those. Eleven observations cannot separate two collinear regressors.

The direction is worth one sentence, and no more. Under the endogeneity account this paper argues for, late money should look *worse* than early money, because committees move resources late into races that have visibly deteriorated, so lateness partly encodes bad news about the recipient. The point estimates run the other way. At these intervals that observation is decoration, not evidence.

The next-cycle placebo, and why the redistricting boundary halves it. Money spent in 2024 cannot have moved a 2022 result, so regressing the cycle-*t* residual on cycle-*(t+1)* independent expenditure should return zero if the contemporaneous association is causal. Washington redrew every congressional district for 2022, so a pair spanning that boundary compares a district label to itself across a redraw and tests nothing about a district's persistent character; those nine pairs are reported

separately and never pooled. That leaves **14 same-era pairs**, on which the placebo returns **+0.878** against a contemporaneous **+1.108** estimated on the same cells — future money apparently "predicting" a past residual at four-fifths the strength of concurrent money, which if precisely estimated would be a serious problem for any causal reading.

It is not precisely estimated. **Deleting one race reverses the placebo's sign** (dropping WA-03's 2022 pair moves it from +0.878 to −0.716), and deleting one race quadruples the contemporaneous comparison (+1.108 to +4.415). Both numbers describe individual contests rather than a relationship. The placebo therefore cannot discriminate, which is the underpowered outcome and not a clean bill of health — it neither corroborates the contemporaneous estimate nor convicts it of confounding.

One incidental result is worth recording because it is cleaner than either regression: across same-era pairs, a district's net independent expenditure correlates with its own next-cycle figure at **+0.031**. Washington's outside money does not persist in a seat. It arrives where a given cycle's contest is expected to be close, and goes elsewhere when it is not — which is the endogeneity mechanism visible directly, without a regression.

What more cycles would and would not fix — now partly answered by the legislative panel. The natural objection to a null on 34 races is that 34 is too few. The state-legislative extension is the closest thing to a test of that objection this record affords — with its scope stated exactly: it holds the **estimator** fixed and multiplies the cells by nearly four, to 129, but under a **related, non-equivalent measurement system** (PDC C-6 rather than FEC Schedule E; three-quarters of the directional dollars electioneering rather than express advocacy; chamber-level House reporting; one scored contest per district-cycle), so it is an external extension, not a pure sample-size experiment. Within that scope the answer is informative: the interval did not narrow and the sign did not settle; it became *specification*-dependent instead, and the standardized comparison above shows why — usable treatment variation, not cell count, is the binding quantity. More cycles would still narrow the interval in 2c, which spans −0.600 to +2.821, and would give the placebo enough same-era pairs to be worth running. They are not freely available. FEC's Schedule-E record extends back well past this panel, but the binding constraint is the *dependent variable*: the residual requires a fundamentals baseline, the baseline requires a Cook-style PVI, and PVI requires presidential results from before the target year. Washington's loaded results begin with the 2016 general, so a 2016 residual is undefined for want of 2012 presidential precinct data — and the Secretary of State's bulk export, which serves 2014 and 2016, returns 404 for 2012. Extending backward is therefore a data-acquisition project against an older and less uniform source, not a re-run of the loader. Extending forward is automatic but slow: November 2026 adds one cycle.

And none of it would make the estimate identified. The panel has no exogenous variation in spending; spending is aimed at expected closeness. A narrower interval around a better-estimated association is still an association. Settling the question needs a design this record cannot supply — a discontinuity, a lottery, or an experiment — and no quantity of FEC backfill produces one.

What it means

Money in Washington elections is **clearly a marker of candidate strength; whether it is *also* a mover of votes, this design does not establish either way.** Those are not mutually exclusive: a strong candidate can attract money and then gain a further point *because* of the spending, and nothing in an observational record decomposes the two. The three manifestations examined here each fall short of identification in its own way: allocation is an exploratory composition cut that predicts nothing out of sample, the forecast model's discarded term is a forecasting result that cannot separate marker from embedded effect, and the directional cross-section's positive slope is a single-district leverage result whose every interval — case-resampled or clustered — spans zero.

Two things follow, and they pull in opposite directions.

The first is a caution against the strong reform claim. "Money buys elections" is not supported by this evidence, and the donor paper's findings should not be read as implying it. What the donor paper establishes is that a narrow, old, geographically concentrated population *funds* campaigns — which matters for whose calls get returned and whose concerns reach an agenda, whether or not it moves a single vote. The companion cut in `cross-state-fec-money.md` §J sharpens this: in Washington's Solid seats the longshot party receives just **4.8%** of House inflow. Money overwhelmingly reaches candidates who were going to win anyway. That is a finding about *access*, not persuasion.

The second is a caution against the strong null. This paper has not shown money does not matter. It has shown that Washington's public record cannot demonstrate that it does, on the vote-moving axis specifically, at the sample sizes available. The **access and agenda-setting channel is entirely untested here** — it is the channel the donor paper's findings actually bear on, and the one most likely to matter. An honest reader should leave with the question open and a clear sense of what it would take to close it.

What this paper does not claim, and limits

- **This is not a causal estimate, and no causal claim is made in either direction.** Every cut is observational with no exogenous variation and no instrument. The +0.60 correlation is exactly what a real causal effect would also produce; the nulls are consistent with a true zero *and* with an effect the data is too thin to detect — and no bootstrap interval in this paper bounds a causal effect, only an observational slope.
- **The allocation null is underpowered and tests the wrong thing twice over.** Coverage is 40 of 129 legislative baseline cells, holdout n runs 15–22, and it tests spend *mix*, not spend *level*. It cannot rule out an effect of simply having more money.
- **The IE panel is n = 34, its sign is not stable, and its point estimate is a leverage result.** The estimate moved from −0.39 to +0.515 when the panel went from one cycle to four, and deleting the single highest-leverage race (WA-08 2018, Cook's distance 0.69) moves it to −0.035. Read the intervals — including the wider district-clustered one — not the coefficient; a paper quoting either sign as a result would be over-reading its own data.
- **State-legislative IE now enters the analysis, and its specification is the live limitation.** An earlier version of this bullet said it was excluded entirely for want of a directional flag. That was wrong about the record rather than about the analysis: the PDC publishes direction in form C-6's C6.3 section, which the loader had not been reading, so the $51.7M panel is in and contributes 129 scorable district-cycles (Finding 3). What remains is that three quarters of the directional dollars are electioneering communication rather than express advocacy and the two run in opposite directions, and that the filing names a candidate's chamber but not their House position — so the point estimate's sign is specification-dependent rather than merely imprecise.
- **Only the persuasion channel is examined.** Access, agenda-setting, candidate entry deterrence, and the effect of money on *who runs in the first place* are all untested and all plausible routes by which money could matter without moving a general-election margin.
- **Coverage gaps favor larger campaigns.** PDC mini-reporting excludes the smallest campaigns, and 38% of expenditure dollars carry no purpose code.

Appendices

Appendix A — The objections, in full

1. "You found +0.60 and then explained it away." The correlation is real and the paper leads with it. The objection has force: the burden of proof for dismissing the largest correlation in the dataset should be high. But nothing here dismisses it, and nothing here is entitled to: what the paper declines to do is *sign* it causally, because the observational record cannot separate selection into money from an effect of money. Allocation carrying no out-of-sample signal and the forecast model finding the term redundant against the baseline are consistent with money tracking a strength that already exists — and equally consistent with a real effect whose information is already embedded in the baselines those tests condition on. The directional cross-section decides nothing in either direction: its positive point estimate rests on one district and every interval around it spans zero.

2. "Absence of evidence is not evidence of absence." Correct, and the paper's title question is answered "cannot tell," not "no." The allocation test is underpowered, the IE interval is too wide to sign, and neither speaks to spend *level* with exogenous variation. This is stated in the body rather than buried, because the alternative — presenting a null as a finding of no effect — would be the more serious error.

3. "The IE slope is just strategic targeting, which you admit." Yes, and that is the point rather than a flaw. Whatever its sign, the association is offered as evidence of *endogeneity*, not of money being harmful or helpful. Spending is aimed at races expected to be close, so the naive correlation cannot be read causally in either direction — which is the paper's central methodological claim, and the reason the sign reversing between drafts changes nothing about the argument.

4. "WA-03 is one race." It is, and it carries no inferential weight. It appears because it is the most extreme case in this panel — the largest independent-expenditure total in five cycles of Washington House races, landing 0.06 points from its fundamentals — and because a reader who suspects the null is an artifact of small money should know what the largest observation looks like. It is not, as an earlier draft claimed, the most IE-saturated House race in the country; it ranks 22nd.

5. "Your baseline is model-derived, so you are testing your own model." Partly true and worth stating plainly. Overperformance and the IE residual are both measured against a fundamentals baseline this project built. That baseline is

validated independently (`safe-seat-washington.md` shows its safe-seat projection matching observed results within a few points), but a reader who rejects the model should read Finding 1's correlation, which uses the neutral PVI baseline, and discount Findings 2b and 2c accordingly.

Appendix B — Data access and provenance

- **Election results.** Certified precinct-level returns from the Washington Secretary of State, aggregated to the seat. Public aggregate records.
- **Candidate finance.** Washington PDC candidate summaries and itemized C4 Schedule-A expenditures (Socrata `tijg-9zyp`), plus FEC candidate summaries for congressional races. Public disclosure records.
- **Independent expenditures.** FEC Schedule E for the **2018–2026 cycles**, carrying support/oppose and district, read with `is_notice=false` so each expenditure is counted once (Appendix C); PDC independent-expenditure records for state legislative races, which carry amounts, races **and direction** — the support/oppose flag sits in form C-6's C6.3 "Identified Entities" section (Finding 3). *(This entry read "for the 2024 cycle" and "no directional flag" until 2026-08-10. Both were true of the extraction, not of the source, and both contradicted Finding 3 in the same document.)*
- **No personal data.** This paper uses candidate- and race-level aggregates only. It touches no voter file and releases no individual record.
- Full provenance and reproduction recipe: Data Sources & Reproducibility.

Appendix C — Methods

- **Overperformance.** `actual_dem_two_party_pct − baseline_dem_pct`, where the baseline is the model's neutral PVI plus down-ballot-drag figure, *not* the full prediction. Positive means the Democrat beat the district's fundamentals. Universe: the backtest grid of 10 congressional and 49 legislative districts across 2018–2024, restricted to the **163 baseline-scorable cells** — cells whose PVI fell back to a default for want of district presidential data are excluded, the same 163 used by the published backtest.
- **A 129 and a 128 appear in this paper, and they are not the same population.** Of the 163 baseline-scorable cells, **129 are legislative** and 34 congressional; that legislative 129 is the denominator in the allocation-coverage limit ("40 of 129") and is the frame Finding 3's state-legislative panel is matched to. Separately, **128 of the 163 carry both-side finance** and are Finding 1's correlation frame — the two counts were equal by coincidence (both 129) until the 2026-08-15 linkage correction removed one wrongly-attached finance pair, and the earlier collision of names is recorded as a withdrawn claim in `who-decides-washington.md`.
- **Fundraising feature.** Raw `log2(D receipts / R receipts)` matched against `candidate_finance`, **scoped to the cell's own office and district** since 2026-08-15: the key is (cycle, last name, first initial) *within* office `H` and the padded district for congressional cells, or offices `SR` / `SS` and the bare district for legislative ones — the same dual-form office/district rule the codebase's finance queries use. The earlier key omitted office and district and took the maximum receipts across all four office codes, which is far weaker entity resolution than the companion donor papers allow themselves; the collision audit (`scripts/diag_finance_linkage_audit.py`) measured the damage at 2 of 326 lookups — one same-key maximum taken from the wrong district, and one candidate credited with a **same-name namesake's** $190K from another chamber. The feature is computed independently of the model's own capped fundraising term.
- **Allocation shares.** Field, media and professional shares of *coded operational* spend per candidate, so the measure is composition rather than level. Cross-cycle holdout fits 2022 and predicts 2024, the same bar used to reject the candidate-quality index earlier in this project.
- **IE residual test.** Net pro-Democratic IE = support-D plus oppose-R minus support-R minus oppose-D, per race, from FEC Schedule E. The dependent variable is the fundamentals-net residual, not the margin. Uncontested races and races without national-cycle data are dropped, leaving 34 of 50 candidate race-cycles scorable.
- **Notice de-duplication.** FEC Schedule E is read with `is_notice=false`, which selects the periodic Form 3X filing and excludes the 24/48-hour Form 24 notice restating the same expenditure. Memo rows (`memo_code='X'`) are excluded as subtotals of money itemized elsewhere. Both exclusions are properties of the summing view, not of the stored table, which retains the notice rows so the reporting lag stays observable.
- **Estimator and intervals, everywhere.** Every slope in this paper — the federal IE regression, all four state-legislative specifications, the placebo pair, and the early/late split — is estimated by **ordinary least squares** (simple bivariate OLS; the early/late split is a two-regressor OLS), and every correlation is Pearson's r. Every "95% bootstrap interval" is a

case-resampling (pairs) percentile bootstrap with 5,000 replicates and a fixed seed, so the intervals are deterministic and reproduce exactly. No heteroskedasticity-robust or weighted variant is used anywhere; at these sample sizes the honest summary is the interval, not the standard-error recipe.

- **Inference threshold.** The IE script refuses to report a slope as inferential below 10 scorable races and prints its data inventory instead. That threshold is now cleared; the behavior is deliberate and should be preserved.
- **Reproduction.** `scripts/diag_overperformance_patterns.py` (Finding 1), `scripts/diag_expenditures_vs_residual.py` (2a), `scripts/diag_ie_vs_margin.py` (2c and Finding 3), `scripts/diag_loser_side_money.py` (the §J longshot-share figure).

Appendix D — Related work

This paper contributes a negative result and a data-availability audit, not a new identification strategy. It sits in a literature that has been circling the same endogeneity problem for five decades:

- **The endogeneity problem itself.** Jacobson, "The Effects of Campaign Spending in Congressional Elections," *American Political Science Review* 72(2) (1978): 469–491 — the founding observation that incumbent spending appears ineffective because incumbents raise and spend most when threatened. Jacobson, "The Effects of Campaign Spending in House Elections: New Evidence for Old Arguments," *American Journal of Political Science* 34(2) (1990): 334–362. The Washington pattern in Finding 2c is this problem reappearing in independent-expenditure data.
- **Designs that address it — and that this data cannot support.** Levitt, "Using Repeat Challengers to Estimate the Effect of Campaign Spending on Election Outcomes in the U.S. House," *Journal of Political Economy* 102(4) (1994): 777–798, doi:10.1086/261954 — differencing out candidate quality using repeated identical match-ups, and finding spending effects far smaller than naive estimates. Gerber, "Estimating the Effect of Campaign Spending on Senate Election Outcomes Using Instrumental Variables," *American Political Science Review* 92(2) (1998): 401–411 — the instrumental-variables approach. Both require panel structure or an instrument this record lacks.
- **The experimental literature.** Kalla & Broockman, "The Minimal Persuasive Effects of Campaign Contact in General Elections: Evidence from 49 Field Experiments," *American Political Science Review* 112(1) (2018): 148–166, doi:10.1017/S0003055417000363 — the best available evidence that general-election persuasion effects are approximately zero on average, which is the prior this paper's nulls are consistent with.
- **Money as information rather than purchase.** Bonica, "Mapping the Ideological Marketplace," *American Journal of Political Science* 58(2) (2014): 367–386, doi:10.1111/ajps.12062 — contributions as a revealed-preference signal about candidates, the reading most consistent with the results here.
- **Access rather than votes.** Kalla & Broockman, "Campaign Contributions Facilitate Access to Congressional Officials: A Randomized Field Experiment," *American Journal of Political Science* 60(3) (2016): 545–558, doi:10.1111/ajps.12180 — the channel this paper explicitly does *not* test, and the one where an experimental effect has actually been demonstrated. Any reader inclined to conclude "money doesn't matter" from this paper should read that one.

Appendix E — The full IE cross-section

Every Washington congressional race with FEC Schedule-E data on disk, 2018–2024 (`scripts/diag_ie_vs_margin.py`). "Residual" is actual minus model-predicted Democratic share; *unscorable* means the race was uncontested or lacked national-cycle data. The 2026 cycle is omitted: it carries 50 flagged rows and no scorable race.

race	net pro-D IE	total IE	residual (pp)	actual margin
cd01 / 18	$0.00M	$0.00M	−4.15	+4.74
cd02 / 18	$0.00M	$0.00M	— *(unscorable)*	—
cd03 / 18	+$1.06M	$3.18M	+2.18	−5.36
cd04 / 18	+$0.03M	$0.03M	−4.62	−25.55
cd05 / 18	+$0.03M	$0.23M	−1.45	−9.46
cd06 / 18	$0.00M	$0.00M	−4.51	+27.77
cd07 / 18	$0.00M	$0.00M	+5.82	+67.12

race	net pro-D IE	total IE	residual (pp)	actual margin
cd08 / 18	+$8.26M	$16.28M	+8.78	+4.84
cd09 / 18	+$0.01M	$0.01M	— *(unscorable)*	—
cd10 / 18	$0.00M	$0.00M	−6.07	+23.09
cd01 / 20	$0.00M	$0.00M	+1.74	+17.25
cd02 / 20	$0.00M	$0.00M	−3.25	+26.48
cd03 / 20	−$1.54M	$3.79M	−0.20	−13.01
cd04 / 20	+$0.01M	$0.01M	−1.21	−32.63
cd05 / 20	−$0.15M	$0.15M	−0.72	−22.82
cd06 / 20	$0.00M	$0.00M	−2.84	+18.90
cd07 / 20	$0.00M	$0.00M	+8.01	+66.37
cd08 / 20	+$0.02M	$0.12M	+0.03	+3.57
cd09 / 20	$0.00M	$0.00M	+7.17	+48.52
cd10 / 20	+$0.76M	$1.38M	— *(unscorable)*	—
cd01 / 22	$0.00M	$0.00M	+4.91	+27.13
cd02 / 22	$0.00M	$0.00M	−3.51	+20.35
cd03 / 22	+$1.45M	$5.73M	+9.16	+0.87
cd04 / 22	−$0.42M	$1.74M	−2.40	−36.05
cd05 / 22	−$0.11M	$0.11M	+1.46	−19.23
cd06 / 22	$0.00M	$0.00M	−0.96	+20.14
cd07 / 22	$0.00M	$0.00M	+9.19	+71.49
cd08 / 22	−$0.20M	$17.47M	+0.38	+6.90
cd09 / 22	$0.00M	$0.00M	+5.59	+43.49
cd10 / 22	$0.00M	$0.00M	−3.22	+14.13
cd01 / 24	$0.00M	$0.00M	+7.59	+26.35
cd02 / 24	$0.00M	$0.00M	— *(unscorable)*	—
cd03 / 24	**+$6.09M**	**$18.61M**	**+0.06**	**+3.92**
cd04 / 24	+$0.99M	$3.35M	— *(unscorable)*	—
cd05 / 24	−$0.09M	$0.23M	+1.55	−21.25
cd06 / 24	+$2.74M	$2.74M	−3.52	+13.67
cd07 / 24	$0.00M	$0.00M	+13.70	+68.35
cd08 / 24	+$0.42M	$0.42M	+6.82	+8.18
cd09 / 24	$0.00M	$0.00M	— *(unscorable)*	—
cd10 / 24	$0.00M	$0.00M	−1.17	+17.35

Thirty-four scorable races. Party attribution is effectively complete — $0.03M of the $75.7M total is unresolvable to a side, so the slope is not a coding artifact. Note how much of the variation sits in races with **no** independent expenditure at all: cd07's

+13.70 residual in 2024 and +9.19 in 2022 both occur at $0 IE, which is a compact illustration of why even thirty-four observations cannot separate money's effect from everything else that varies across districts and cycles.

Note also the two largest totals in the panel, cd08 in 2018 ($16.28M) and 2022 ($17.47M), which bracket WA-03's 2024 $18.61M. All three landed within nine points of their fundamentals and two of the three within one point.

End note — data, reproduction, and series

```
# Finding 1 — money vs overperformance across 163 race-cycles:
python scripts/diag_overperformance_patterns.py

# Finding 2a — spend allocation vs residual, cross-cycle holdout:
python scripts/diag_expenditures_vs_residual.py

# Finding 2c — federal IE vs fundamentals-net residual (34 WA U.S. House cells).
# In the public repo this reads the pinned panel and recomputes the slope, the
# bootstrap interval and Pearson r from it; the residual itself needs the
# forecast model, which is not published.
python scripts/diag_ie_vs_margin.py

# Finding 3 — the same design on WA legislative races (129 cells, PDC C6.3).
# Load the direction data first; it is a separate section of form C-6:
python main.py load --district ld01 --pdc-ie-targets
python scripts/diag_pdc_ie_vs_margin.py

# The C6.3 reconciliation the legislative panel rests on (needs no database):
python scripts/diag_pdc_c63_reconciliation.py

# The longshot-share figure quoted in "What it means":
python scripts/diag_loser_side_money.py
```

All inputs are public records: certified election returns, WA PDC disclosure, and FEC bulk and API data. See `data-sources-and-reproducibility.md` for the full source ledger.

This is Paper #4 of the electoral-health series: `who-decides-washington.md` (who votes), `safe-seat-washington.md` (whether the vote decides anything), `donor-class-and-the-electorate.md` (who funds it), and `cross-state-fec-money.md` (the four-state money layer).